

Kushal Trivedi

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EDUCATION

Master of Science, Data Science December 2026
Arizona State University, Tempe, AZ GPA: 3.85

Bachelor of Engineering, Computer Science June 2024
New L.J. Institute of Engineering & Technology, India GPA: 3.88

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, JavaScript, Django, Flask, FastAPI, React

ML & Data Engineering: PyTorch, TensorFlow, scikit-learn, OpenCV, Transformers, RAG, ETL pipelines, MLflow, DVC

Databases & Storage: PostgreSQL, MySQL, SQLite, MongoDB

DevOps & Cloud: AWS (S3, EC2), Docker, GitHub Actions (CI/CD), Git/GitHub

EXPERIENCE

Software Developer Intern, Praxware Technologies Jan 2024 – June 2024

- Built and automated legal-services workflows across 4 role-based portals (client, lawyer, admin, public) in Python, Django, JavaScript, and MySQL, architecting **18 relational models** and **90+ REST/page endpoints**, each with independent authentication and secure case-document uploads.
- Integrated **3 third-party services** – Razorpay, Jitsi Meet, and ReportLab – to enable online subscription payments, live video consultations, and automated PDF case-summary generation.

PROJECTS

Sentiment Intelligence Platform Aug 2026

- Fine-tuned a DistilBERT transformer with partial layer freezing on 50K stratified Yelp reviews for 5-class sentiment classification, improving F1 by **9.3%** over a TF-IDF baseline and reaching macro F1 of 0.63 and ROC-AUC of 0.90; tracked experiments with MLflow and data versions with DVC.
- Deployed real-time predictions through a FastAPI backend exposing 4 REST endpoints for health checks and single/batch prediction, containerized with Docker, explained via SHAP token attribution, and monitored with a custom data-drift detector.

Glacier Melt Prediction & Climate Impact Analysis Jun – Aug 2026

- Engineered a 9-process analytics pipeline over 15+ years of NASA satellite imagery of the Gangotri Glacier, calibrating pixel-spectral glacier-coverage, ocean-temperature, and sea-level-anomaly indices with statistical shadow and outlier detection to uncover a **-0.79%/yr** glacier-coverage retreat trend.
- Built an interactive Flask dashboard with click-to-satellite-image inspection, year-range filtering, and 13 charts, surfacing multi-signal correlations up to **Pearson $r = 0.39$** ; added an AI Insights module powered by the Groq API that auto-generates narrative explanations for all charts in a single batched call.

Controllable Face Image Generation Apr – May 2026

- Designed and implemented a ControlNet-augmented Stable Diffusion inference pipeline with Canny-edge structural conditioning at 512x512 resolution and auto-generated multi-attribute text prompts, preserving facial identity during prompt-driven attribute edits – the individually-owned component of a 5-member team project benchmarking DCGAN, ACGAN, and ControlNet on a 5,000-image CelebA subset.
- Validated that the pipeline generalizes rather than memorizes by running the full 5,000-image generation on a Google Colab T4 GPU, reaching a mean CLIP ViT-B/32 alignment score of **23.0 ± 3.0** with a near-zero train/val/test gap.

Book Recommendation System Mar – Apr 2025

- Built the data-preprocessing and sparse-matrix construction pipeline using SciPy's `csr_matrix` with LIBSVM export for a 4-person collaborative-filtering project on the Book-Crossing dataset, encoding a 105,283-user x 340,556-book ratings matrix.
- Generated **210,000+** personalized top-5 book recommendations by computing cosine similarity across the top-10 nearest neighbors per user and ranking candidates via similarity-weighted rating estimates, powering the team's recommendation engine.

NYC Taxi Trip Data Mining & Fare Prediction Jan – Mar 2025

- Cleaned and refined 11.2M+ NYC taxi records to **8.5M+ valid rows** as part of a 4-person course project, engineering temporal features (trip duration, pickup hour, weekday, month) for spatio-temporal demand analysis.
- Benchmarked 5 regression models – Random Forest, Gradient Boosting, Ridge, SVR, and Linear Regression – reaching **$R^2 \approx 0.93$** and a 3% RMSE reduction over baseline; clustered pickup zones with K-Means ($k=5$, silhouette 0.385) and flagged anomalous trips with Isolation Forest at 2% contamination.